Computational Project No. 2

1. Create a code for solving fixed-source transport problems in one-dimensional slab geometry by means the QD method using Algorithm 1 and:

   - the lumped linear discontinuous (LLD) scheme for discretization of the high-order transport equation,
   - the finite volume (FV) scheme of the second-order accuracy for discretization the low-order QD (LOQD) equations,
   - the discrete prolongation operator of the QD method on the spatial mesh which defines mapping of the grid functions of the scalar flux from the discretized LOQD equations to the grid functions of the discrete scalar flux in the scattering term of the LLD scheme for the high-order transport equation.

Algorithm 1: Iteration scheme of solving the transport problem by means of the QD method

$s = 0$
$\rho_\phi^{(1)} = \frac{1}{2}$
set $E^{(1/2)} = \frac{1}{3}$, $C_L^{(1/2)} = -\frac{1}{2}$, $C_R^{(1/2)} = \frac{1}{2}$, $\phi^{(0)} = 1$
while $||\phi^{(s)} - \phi^{(s-1)}||_\infty > \varepsilon_\phi\left(\frac{1}{\rho_\phi^{(s)}} - 1\right)$ **do**
  $s = s + 1$
  if $s > 1$ **then**
    solve the high-order transport problem to calculate $\psi^{(s-1/2)}$ using $\phi^{(s-1)}$
    compute $E^{(s-1/2)}$, $C_L^{(s-1/2)}$ and $C_R^{(s-1/2)}$ using $\psi^{(s-1/2)}$
  solve the low-order QD problem for $\phi^{(s)}$ and $J^{(s)}$ using $E^{(s-1/2)}$, $C_L^{(s-1/2)}$ and $C_R^{(s-1/2)}$ as coefficients of the LOQD equations
  if $s > 2$ **then**
    calculate $\rho_\phi^{(s)} = \frac{||\phi^{(s)} - \phi^{(s-1)}||_\infty}{||\phi^{(s-1)} - \phi^{(s-2)}||_\infty}$
  else
    $\rho_\phi^{(s)} = \frac{1}{2}$

2. Solve the tests problems below to compute numerical solution including grid functions of

$$\psi(x,\mu)\,, \quad \phi(x)\,, \quad J(x)\,, \quad E(x)\,, \quad C_L\,, \quad C_R \tag{1}$$

and metrics of transport effects

$$M^E(x) = \left|\frac{1}{3} - E(x)\right|, \quad M^{E'}(x) = \left|\frac{dE}{dx}(x)\right|. \tag{2}$$

- **Test A**
  - There are two spatial regions in the problem:
    - $\mathcal{A}_1$: $x \in [0, 1]$, $\sigma_t = 2$, $\sigma_s = 0$, $\sigma_f = 0$, $q = 0$,
    - $\mathcal{A}_2$: $x \in [1, 11]$, $\sigma_t = 100$, $\sigma_s = 100$, $\sigma_f = 0$, $q = 0$.
  - The left boundary conditions is given by $\psi(0, \mu) = 1$ for $\mu > 0$.
  - The right boundary condition is vacuum.
  - The spatial mesh in the subdomains is defined with the following intervals:
    - $\mathcal{A}_1$: $\Delta x = 0.1$,
    - $\mathcal{A}_2$: $\Delta x = 1$.
  - Use the double S_4 Gauss-Legendre quadrature set.
  - $\varepsilon_\phi = 10^{-10}$.
- **Test B**
  - There are two spatial regions in the problem:
    - $\mathcal{A}_1 : x \in [0, 10]$, $\sigma_t = 1$, $\sigma_s = 0.5$, $\sigma_f = 0$, $q = 0.5$,
    - $\mathcal{A}_2 : x \in [10, 30]$, $\sigma_t = 10$, $\sigma_s = 9.9$, $\sigma_f = 0$, $q = 0.01$.
  - The boundary conditions are vacuum.
  - The spatial mesh in the subdomains is defined with the following intervals:
    - $\mathcal{A}_1$: $\Delta x = 1$,
    - $\mathcal{A}_2$: $\Delta x = 0.5$.
  - Use the double S_4 Gauss-Legendre quadrature set.
  - $\varepsilon_\phi = 10^{-10}$.
- **Test C**
  - There are two spatial regions in the problem:
    - $\mathcal{A}_1 : x \in [0, 10]$, $\sigma_t = 1$, $\sigma_s = 0.5$, $\sigma_f = 0$, $q = 0.5$,
    - $\mathcal{A}_2 : x \in [10, 30]$, $\sigma_t = 10$, $\sigma_s = 9.9$, $\sigma_f = 0$, $q = 0$.
  - The boundary conditions are vacuum.
  - The spatial mesh in the subdomains is defined with the following intervals:
    - $\mathcal{A}_1$: $\Delta x = 1$,
    - $\mathcal{A}_2$: $\Delta x = 0.5$.
  - Use the double S_4 Gauss-Legendre quadrature set.
  - $\varepsilon_\phi = 10^{-10}$.

- **Test D**
  - This is a one-region problem:
    (a) $\mathcal{A} : x \in [0,1]$, $\sigma_t = \frac{1}{\nu}$, $\sigma_a = \nu$, $\sigma_f = 0$, $q = \nu$,
  - (a) $\nu = 10^{-2}$, (b) $\nu = 10^{-4}$
  - The boundary conditions are vacuum.
  - The spatial mesh is uniform with $\Delta x = 0.1$.
  - Use the double S_4 Gauss-Legendre quadrature set.
  - $\varepsilon_\phi = 10^{-10}$.

3. In your report you need include

   (a) flowcharts of algorithms,

   (b) the discretization method for the system of equations of the QD method presenting

      i. the LLD scheme for discretization of the high-order transport equation,

      ii. the FV scheme for discretization the LOQD equations,

      iii. mapping of the grid functions of the scalar flux from the discretized LOQD equation to the grid functions of the discrete scalar flux in the scattering term of the LLD scheme for the high-order transport equation.

   (c) files with your code,

   (d) the set of numerical results in the following form:

   - plot the cell-average values of angular flux, $\psi_{i,m}$, versus (x_i, μ_m) (note that it is a 3D plot),
   - plot the cell-average values of scalar flux, ϕ_i, from the low-order QD problem versus x_i and
   - plot the cell-edge values of scalar flux, $\phi_{i+1/2}$, from the low-order QD problem versus $x_{i+1/2}$,
   - plot of the cell-edge values of current, $J_{i+1/2}$, from the low-order QD problem versus $x_{i+1/2}$,
   - values of C_L and C_R
   - plot of the cell-average values E_i versus x_i,
   - plot of the cell-edge values $E_{i+1/2}$ versus $x_{i+1/2}$,
   - plot of the cell-average values of the metric M^E of transport effects

     $$M_i^E = \left| \frac{1}{3} - E_i \right| \tag{3}$$

     versus x_i,
   - plot of the cell-average values of another metric of transport effects

     $$M_i^{E'} = \left| \frac{E_{i+1/2} - E_{i-1/2}}{\Delta x_i} \right| \tag{4}$$

     versus x_i,

- plot of convergence histories

$$||\phi^{(s)} - \phi^{(s-1)}||_\infty \tag{5}$$

  versus s on logarithm scale
- plot of estimation of spectral radius

$$\rho^{(s)} = \frac{||\phi^{(s+1)} - \phi^{(s)}||_\infty}{||\phi^{(s)} - \phi^{(s-1)}||_\infty} \tag{6}$$

  versus s
- the number of iterations.

Problem	Test A	Test B	Test C	Test D (a)	Test D (b)	total	%
Points							
Maximum	10	10	10	10	10	50	